

Foundation of Blockchain Technology (IS501)

Pre-Reading Material – Experiment 1

Blockchain Architecture

Estimated Reading Time: 20–25 minutes

Estimated Reading Time: 15–20 minutes

Why This Topic Matters

Blockchain is more than just cryptocurrency—it is a secure and decentralized system for storing and sharing data. Understanding its architecture helps explain how transactions are verified, how blocks are connected, and why blockchain is considered transparent, secure, and tamper-resistant. This knowledge forms the foundation for studying consensus mechanisms, smart contracts, and blockchain applications in later lectures.

Learning Outcomes

After completing this pre-reading and attending the lecture, you should be able to:

- Explain the basic architecture of a blockchain.
- Identify the major components of a blockchain network.
- Describe the structure of a block.
- Explain the role of cryptographic hash functions.
- Differentiate between centralized and decentralized architectures.
- Describe how transactions are added to the blockchain.

Activate Your Prior Knowledge

1. Why is Google Drive reliable even if one server fails?

Think about how storing copies of data on multiple computers improves availability and reliability.



2. Why does changing one character in a PDF create a different file?

Consider how computers identify files and detect changes using digital fingerprints.

3. Why do banks maintain transaction records?

Think about why maintaining a permanent transaction history is important for security and accountability.

4. Can multiple computers maintain the same database without a central server?

Reflect on how a distributed system differs from a centralized system.

Prerequisite Concepts

Before attending this lecture, you should be familiar with:

- Basic concept of blockchain.
- Difference between centralized and distributed systems.
- Basic computer networks.
- Digital transactions and databases.

No prior knowledge of cryptography or programming is required.

Core Concepts

1. Blockchain Architecture

Blockchain architecture is the overall structure of a blockchain network. It defines how transactions, blocks, nodes, and the distributed ledger work together to maintain a secure system.

2. Transaction

A transaction is any activity recorded on the blockchain, such as transferring cryptocurrency or updating digital information. Transactions are verified before being added to a block.

3. Block

A block is a collection of verified transactions stored together. Each block contains transaction data and information that links it to the previous block.

4. Block Header

The block header stores important information such as the previous block hash, timestamp, nonce, and Merkle root. It uniquely identifies the block within the blockchain.

5. Hash Function

A cryptographic hash function converts input data into a fixed-length hash value. Even a small change in the input produces a completely different output, ensuring data integrity.

6. Previous Block Hash

Each block stores the hash of the previous block. This creates a continuous chain of blocks and makes unauthorized modifications easy to detect.

7. Node

A node is a computer connected to the blockchain network. It stores a copy of the blockchain and participates in verifying and sharing transactions.

8. Distributed Ledger

A distributed ledger is a shared database maintained by all participating nodes. Every node keeps the same copy of the blockchain, improving transparency and reliability.

9. Centralized vs Decentralized Network

A centralized system is controlled by a single authority, whereas a decentralized blockchain network distributes control among multiple participating nodes.

10. Blockchain Network

A blockchain network is formed by interconnected nodes that collectively validate, store, and synchronize blockchain data without relying on a central authority.

11. Data Integrity

Data integrity means that blockchain records remain accurate and unchanged after validation. Cryptographic hashing helps detect any unauthorized modifications.

12. Immutability

Immutability means that once information is recorded on the blockchain, it cannot be easily altered or deleted. This makes blockchain suitable for maintaining trustworthy records.



Guided Reading Questions

1. Why is blockchain called a distributed ledger?

Identify how multiple nodes maintain identical copies of the blockchain instead of relying on a central database.

2. What information is stored inside a block?

Observe the different components of a block and understand their purpose in maintaining blockchain security.

3. Why is the previous block hash important?

Think about how linking blocks together protects the blockchain from unauthorized modifications.

4. What is the role of a node in a blockchain network?

Consider how nodes participate in storing, validating, and sharing blockchain information.

5. Why is blockchain considered tamper-resistant?

Look for the relationship between cryptographic hashing and block linking that makes blockchain highly secure.

Reflection Activity

In **150–200 words**, explain how blockchain architecture differs from a traditional centralized database. Discuss the roles of transactions, blocks, nodes, and cryptographic hashing in maintaining security and reliability.

Self-Assessment

Answer the following before class:

1. What is blockchain architecture?
2. Define a blockchain transaction.
3. What is the purpose of a block?



4. What is a cryptographic hash?
5. Why does every block store the previous block hash?
6. What is a blockchain node?
7. What is a distributed ledger?
8. Differentiate between centralized and decentralized networks.
9. What is meant by immutability?
10. Why is blockchain considered secure?

